

A Static-Universe Two-Component Scattering Interpretation of DES-SN5YR Supernovae Redshift, Temporal Broadening, and the Hubble Diagram

T. McSheery

PhaseSpace Inc., San Leandro, CA, USA

TracyMcSheery@gmail.com

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Abstract

We test a static-geometry interpretation of the Dark Energy Survey five-year Type Ia supernova data (DES-SN5YR) in which cosmological propagation is represented by two distinct photon–vacuum interaction channels rather than metric expansion. The first is a coherent forward channel: photon frequency decreases without angular diffusion and, if phase-cycle count is preserved as the carrier frequency evolves, the arrival envelope obeys $\Delta t_{\text{obs}}/\Delta t_{\text{em}} = \nu_{\text{em}}/\nu_{\text{obs}} = 1+z$. Thus the precise DES measurement $b = 1.003 \pm 0.005$ (stat) ± 0.010 (sys) in $\Delta t_{\text{obs}} \propto (1+z)^b$ is a required consistency condition for the coherent branch, but is not by itself unique evidence for metric expansion. The same frequency scaling maps a Planck-like spectrum to a lower effective temperature through Wien displacement. The second channel is non-forward and chromatic, with a phenomenological cross-section scaling $\sigma_{\text{nf}} \propto E^{1/2} \propto \lambda^{-1/2}$; it removes flux from the coherent beam and predicts a small band-ordered rise/decay asymmetry.

The resulting one-parameter static distance law is nearly degenerate with flat Λ CDM on the released DES-SN5YR Hubble diagram: with the official DES $z_{\text{HD}}/z_{\text{HEL}}$ convention and full released covariance, the pairwise statistic is $\Delta\chi^2 = +1.1$ in favor of Λ CDM. Reconstructing the released vector before the event-level BBC magnitude correction changes the same statistic to $+12.8$, so the total BBC layer carries $L_{\text{BBC}} = -11.7$ units of pairwise leverage. Existing DES simulation tests indicate that the specifically reference-cosmology-dependent portion of BBC is small over an amplitude range larger than the present probe’s scalar deviation, but a matched non-FLRW reference-input rerun has not yet been performed. A prior SALT-free transfer fit to 3,882 DES light curves obtained 0.40-mag scatter and the predicted sign of the weak chromatic trend, but its time-domain implementation predates the corrected $b = 1$ coherence requirement and is therefore treated here as partial consistency, not as a completed time-domain validation.

The central claim is consequently narrow and falsifiable: DES-SN5YR does not presently force an accelerating metric interpretation over this specific static two-channel propagation law. The decisive tests are (i) an observer-frame multiband measurement of the predicted chromatic rise/decay asymmetry and (ii) a matched DES bias-correction/forward-model analysis using the static law as the reference propagation model.

1 The question being tested

The supernova evidence for cosmic acceleration is often presented as the conjunction of two observations: distant Type Ia supernovae are dimmer than a matter-only expanding model predicts,

and their light curves broaden approximately as $(1 + z)$. In the standard interpretation, both are consequences of metric expansion, with the residual distance–redshift curvature requiring a late-time accelerating component.

This paper asks a different question. Suppose the large-scale spatial geometry is static and photon propagation through the vacuum is not perfectly linear. Can one propagation model reproduce the observed spectral redshift, the measured temporal broadening, and the DES Hubble-diagram shape without introducing an accelerating scale factor?

Thesis invariant. A static geometry with two distinct photon-scattering channels can reproduce the three supernova observables usually assigned to expansion—achromatic redshift, approximately $(1 + z)$ temporal broadening, and the luminosity-distance curve—without dark-energy acceleration. DES-SN5YR does not presently distinguish this static propagation law from flat Λ CDM on the released Hubble diagram; the discriminating measurements are the predicted chromatic time-domain residue and a matched non-FLRW survey-simulation rerun.

The word *static* is used here in a deliberately limited sense: there is no cosmological scale-factor expansion. It does not require the effective electromagnetic response of the vacuum to be microscopically time-independent. That distinction is load-bearing because a purely stochastic per-photon “tired-light” process would preserve the macroscopic arrival interval and predict $b \simeq 0$, which DES excludes. The static hypothesis tested here is instead a *coherent* propagation branch in which the carrier frequency and the pulse envelope evolve together.

The model has two observationally distinct channels. Keeping them separate is what avoids the classical tired-light contradictions: the mechanism that shifts spectral lines is not the mechanism that removes photons from the image-forming beam.

Table 1: Observational roles of the two propagation channels.

Channel	Phenomenology	Primary observable consequence
Coherent forward	Direction-preserving fractional energy/frequency change; phase coherence required	Achromatic redshift; if phase cycles are conserved, achromatic $(1 + z)$ arrival-envelope broadening
Non-forward chromatic	Flux removed from the coherent beam with $\sigma_{\text{nf}} \propto E^{1/2}$	Saturating dimming term and a weak wavelength-ordered rise/decay asymmetry

2 Why nonlinear photon propagation is a physical question

Classical electrodynamics is linear in vacuum, but quantum electrodynamics is not. The Euler–Heisenberg effective action predicts field-dependent electromagnetic interactions, and strong-field experiments have directly observed nonlinear QED processes. SLAC E-144 observed nonlinear Compton scattering and multiphoton electron–positron production; the 1997 E-144 result was described as the first laboratory evidence for inelastic light-by-light scattering involving only real photons (Burke et al., 1997; Bamber et al., 1999).

E-144 does *not* demonstrate the cosmological process proposed here: its fields, energies, and interaction rates are completely different. Its relevance is narrower. It removes the premise that

real-photon propagation through the quantum vacuum is fundamentally incapable of nonlinear interaction. The cosmological question is quantitative: whether an effective weak, coherent interaction accumulated over gigaparsec paths can have the required scaling while satisfying astronomical limits on chromaticity, image sharpness, polarization, and energy conservation.

We use “vacuum dielectric” as an effective-medium description, not as a return to a mechanical luminiferous aether. Operationally, the model requires an electromagnetic response that admits a coherent forward amplitude and a much smaller non-forward channel. The observations constrain those amplitudes directly.

3 The coherent branch: redshift and temporal broadening

3.1 Achromatic frequency loss and distance

Let the coherent forward channel produce a constant fractional photon-energy loss per unit path,

$$\frac{dE}{dD} = -\frac{K}{c}E. \quad (1)$$

Since $E = h\nu$, the same equation applies to frequency. Integration gives

$$1 + z \equiv \frac{\nu_{\text{em}}}{\nu_{\text{obs}}} = \exp(KD/c), \quad D(z) = \frac{c}{K} \ln(1 + z). \quad (2)$$

The fractional shift is independent of the initial frequency. Spectral features are therefore translated by the same factor rather than selectively stripped from the spectrum. This is the essential distinction between an achromatic forward drag and classical material scattering.

3.2 What DES time broadening actually constrains

White et al. measured the duration scaling of 1,504 DES Type Ia supernovae over approximately $0.1 \lesssim z \lesssim 1.2$, obtaining (White et al., 2024)

$$\Delta t_{\text{obs}} = \Delta t_{\text{em}}(1 + z)^b, \quad b = 1.003 \pm 0.005 \text{ (stat)} \pm 0.010 \text{ (sys)}. \quad (3)$$

This is a precise observation. A static model that predicts no broadening is excluded.

The coherent scattering hypothesis makes a sharper statement than old tired light. Let N be the number of phase cycles carried by a coherent pulse. If the forward interaction evolves the carrier without creating or destroying phase cycles,

$$N = \nu_{\text{em}} \Delta t_{\text{em}} = \nu_{\text{obs}} \Delta t_{\text{obs}}. \quad (4)$$

Then

$$\frac{\Delta t_{\text{obs}}}{\Delta t_{\text{em}}} = \frac{\nu_{\text{em}}}{\nu_{\text{obs}}} = 1 + z, \quad (5)$$

so $b = 1$ follows exactly. No metric expansion appears in the derivation. What matters is coherence.

Equation (5) is not a proof that the proposed vacuum interaction preserves cycles. It is the condition imposed on the microscopic theory by the DES measurement. A stochastic sequence of independent energy-loss events need not satisfy it and is observationally rejected. In this sense the DES time-dilation measurement selects the coherent branch of a static scattering theory rather than automatically selecting expansion over every possible static theory.

3.3 Wien displacement as the spectral counterpart

For a Planck-like source spectrum, an achromatic frequency mapping $\nu \rightarrow \nu/(1+z)$ preserves the spectral form while shifting its characteristic temperature,

$$T_{\text{eff}} = \frac{T_{\text{em}}}{1+z}, \quad (6)$$

which is equivalent to Wien's displacement law $\lambda_{\text{max}}T = b_{\text{W}}$ with

$$\lambda_{\text{max,obs}} = (1+z)\lambda_{\text{max,em}}. \quad (7)$$

Thus the wavelength displacement, color-temperature evolution, and the cycle-conserving temporal stretch are different observational projections of the same frequency scaling. Wien displacement does not by itself manufacture the achromatic bulk $b = 1$ envelope; rather, it specifies how the evolving spectrum moves through fixed observer filters while Eq. (5) controls the bulk temporal scale.

4 The chromatic branch: dimming and a discriminating asymmetry

The second interaction channel removes photons from the coherent beam. In the current phenomenological form its non-forward cross-section scales as

$$\sigma_{\text{nf}}(E) \propto E^{1/2} \propto \lambda^{-1/2}. \quad (8)$$

Combining Eq. (8) with the coherent energy evolution $E(z) = E_0/(1+z)$ gives a saturating path integral,

$$\tau(z) = \eta \left[1 - (1+z)^{-1/2} \right], \quad (9)$$

where η is an effective opacity amplitude in the present supernova comparison. The functional form is the important part of the DES test; the microscopic normalization remains an open calculation.

The non-forward channel is explicitly chromatic. Because the rising phase of a Type Ia supernova is hotter and bluer than the later decline, Eq. (8) preferentially attenuates the leading edge. The predicted signature is not a second bulk time-dilation factor. It is a small rise/decay asymmetry that strengthens toward bluer observed bands and with accumulated path length.

A template-level calculation in the broader QFD program gives a registered differential-extinction scale of approximately 0.5–3.4 mmag per band over $0.1 \lesssim z \lesssim 1$, with an example $g : r : i : z$ ordering of approximately 1.00 : 0.78 : 0.45 : 0.10 at $z = 0.5$. The implied contamination of a single fitted width exponent is at the few- 10^{-3} level. These numbers are not yet a survey-level detection claim: the dust/circumstellar-medium systematics floor and a fitter-grade observer-frame estimator remain to be completed. They are included because they turn the chromatic channel into a direct falsification target.

5 From the two channels to the supernova Hubble diagram

In static Euclidean geometry the inverse-square flux dilution contributes D^{-2} . The coherent redshift reduces the energy per photon by $(1+z)^{-1}$. If the same coherent process produces the observed arrival-envelope dilation, the photon arrival rate contributes another $(1+z)^{-1}$. Ignoring the non-forward loss for the moment,

$$F_{\text{obs}} \propto \frac{1}{D^2(1+z)^2}, \quad (10)$$

which is observationally equivalent to an effective luminosity distance

$$D_{L,0}(z) = (1+z)D(z) = (1+z)\frac{c}{K}\ln(1+z). \quad (11)$$

The chromatic/non-forward channel adds the saturating magnitude curvature. In the convention used by the companion DES analysis, the probe law is

$$\mu_{\text{static}}(z_{\text{HD}}, z_{\text{HEL}}) \propto 5 \log_{10} \left[(1+z_{\text{HEL}}) \frac{c}{K} \ln(1+z_{\text{HD}}) \right] \quad (12)$$

$$+ \frac{5}{\ln 10} \eta \left[1 - (1+z_{\text{HD}})^{-1/2} \right] + \mathcal{M}, \quad (13)$$

where $\mathcal{M}$ absorbs the absolute-magnitude/distance-scale zero point. The $z_{\text{HEL}}/z_{\text{HD}}$ assignment follows the official DES likelihood convention so that the comparison is not driven by a redshift-frame mismatch.

Equation (13) contains one fitted shape parameter, η , just as flat Λ CDM contains one fitted shape parameter, Ω_m , in the pairwise experiment below.

6 DES-SN5YR results

6.1 Released Hubble diagram: near degeneracy

Using the public DES-SN5YR v1.2 distance vector, the full released STAT+SYS covariance, the official redshift convention, and analytic profiling of the additive magnitude offset, the companion reproduction analysis obtains

$$\Delta\chi^2 \equiv \chi_{\text{static}}^2 - \chi_{\Lambda\text{CDM}}^2 = +1.1 \quad (14)$$

for the released BBC-corrected vector. The best tabulated values are $\Omega_m = 0.352$ and $\eta = 0.297$. Retaining all 1,829 released entries instead of the default 1,768 diagnostic sample changes no quoted headline number.

This is the first central empirical result: the released DES Hubble diagram by itself has essentially no power to distinguish the one-parameter static propagation law of Eq. (13) from one-parameter flat Λ CDM.

6.2 What the BBC correction changes

The released DES distance moduli are not raw flux measurements. They include the BEAMS with Bias Corrections (BBC) layer derived from survey simulations (Kessler & Scolnic, 2017; Vincenzi et al., 2024). Empirically reconstructing the distance vector before the event-level BBC magnitude correction changes the pairwise statistic from +1.1 to +12.8 in favor of Λ CDM. Defining

$$L_{\text{BBC}} \equiv \Delta\chi_{\text{released}}^2 - \Delta\chi_{\text{preBBC}}^2, \quad (15)$$

gives

$$L_{\text{BBC}} = -11.7, \quad (16)$$

roughly eleven times the released discrimination and directed *toward* the static probe. The sign varies for other distance-law probes, so this is leverage, not evidence that BBC is partisan.

This bracket does not isolate the reference cosmology used to generate the BBC simulations. Camilleri et al. directly tested that dependence with alternative FLRW reference cosmologies and

found small recovered-parameter shifts even for reference histories far from baseline (Camilleri et al., 2024). The released-best-fit static probe differs from fitted flat Λ CDM by at most about 0.055 mag after offset removal, below the roughly 0.15-mag scalar amplitude spanned by those tests. That comparison favors the incumbent DES validation at the tested amplitude. The unresolved point is functional rather than merely scalar: the DES validation simulations remain FLRW-family reference models, whereas Eq. (13) is a non-FLRW propagation law. A matched reference-input rerun is therefore the direct completion.

6.3 Raw multiband photometry: what has and has not been shown

DES publicly releases the underlying scene-modeling light curves and ancillary products (Sánchez et al., 2024). A SALT-free transfer-function analysis in the broader project fitted 3,882 supernova light curves using only per-event peak time and amplitude, obtaining an RMS distance scatter of approximately 0.40 mag compared with 0.39 mag for unstandardized SALT peak magnitudes. In that analysis the r, i, z residual trends were each below approximately 0.15 mag over the sampled redshift range, and a low-redshift $g - z$ excess had the sign expected from a weak $\lambda^{-1/2}$ erosion term.

That result is useful but not the final time-domain test. The implementation inherited an earlier $(1+z)^{1/3}$ envelope factor that has since been rescoped to photon coherence/linewidth physics. The DES measurement in Eq. (3) forces the macroscopic envelope exponent to $b = 1$ for any viable version of the static model. The raw-light-curve pipeline must therefore be rerun with the coherent $b = 1$ envelope and a dedicated observer-frame rise/decay estimator before the two-channel time-domain prediction can be called validated.

Table 2: Present evidentiary status of the DES tests.

Observable	Present result	Interpretation
Bulk temporal broadening	$b = 1.003 \pm 0.005 \pm 0.010$ (DES)	Required by coherent cycle-conserving branch; excludes non-dilating stochastic branch
Released Hubble diagram	$\Delta\chi^2 = +1.1$ vs flat Λ CDM	Near degeneracy; DES corrected moduli do not strongly discriminate
BBC leverage	$L_{\text{BBC}} = -11.7$	Processing layer is much larger than final pairwise margin; not attribution to reference cosmology
Raw SALT-free photometry	0.40 mag RMS on 3,882 SNe; weak chromatic sign consistent	Partial consistency only; current $b = 1$ observer-frame refit still required
Chromatic rise/decay asymmetry	Template-level amplitude registered	Decisive direct discriminator; survey-level measurement open

7 Why this is not classical tired light

Zwicky-type tired-light mechanisms failed because a single scattering process was asked to do incompatible jobs: shift frequency, preserve sharp images, remain achromatic, and explain temporal broadening. The present hypothesis separates those jobs.

1. The redshift channel is coherent and forward. A viable microscopic interaction must preserve direction and phase sufficiently well to satisfy astronomical image sharpness and Eq. (5).
2. The flux-loss channel is non-forward and chromatic. Its wavelength dependence is therefore not confused with the achromatic spectral-line shift.
3. The observed $(1 + z)$ broadening is accepted as data, not denied. It is used to reject the incoherent static branch and select the cycle-conserving coherent branch.
4. The two models make a residual time-domain prediction that differs: ideal metric dilation is symmetric and achromatic, whereas the two-channel model adds a small wavelength-ordered rise/decay distortion.

Thus “static” does not mean “no observed dilation.” It means that the measured dilation is attributed to coherent frequency evolution of the propagating signal rather than to the stretching of the spatial metric.

8 Falsification program

The model is useful only if the remaining experiments can kill it. Four tests are particularly direct.

8.1 Observer-frame chromatic timing test

Fit the DES g, r, i, z flux time series directly in observer time, allowing one common bulk exponent b and a band-dependent rise/decay asymmetry. The static model requires $b \simeq 1$ *and* predicts the residual asymmetry to increase toward bluer bands with the registered ordering. A null result below the physical amplitude range falsifies the chromatic non-forward channel.

8.2 Matched BBC reference-input rerun

Generate the DES bias-correction simulations with Eq. (13), or its underlying transfer law, as the reference propagation model and compare the resulting correction with the nominal FLRW-reference correction. This directly measures the currently unisolated reference-model component of L_{BBC} .

8.3 Coherence and image-quality constraints

The forward interaction must change frequency without producing measurable angular diffusion. Any derived microscopic cross-section must therefore satisfy simultaneous constraints from spectral-line achromaticity, diffraction-limited imaging of high-redshift sources, and polarization transport. Failure on any one of these is fatal to the forward channel.

8.4 Microscopic normalization

E-144 establishes nonlinear QED as a real phenomenon, not the required cosmological rate. The theory must derive the effective interaction strength in the low-field astrophysical regime and connect it to the fitted supernova opacity without choosing conventions after the fact. Until that normalization is closed, Eq. (9) is a phenomenological law with a theoretically motivated exponent, not a first-principles prediction of its amplitude.

9 Discussion

The important logical distinction is between an observation and its usual interpretation. DES measures an approximately $(1 + z)$ temporal broadening. That measurement excludes a non-dilating

static model. It does not, by itself, exclude a coherent static propagation law that predicts the same broadening through Eq. (5). Likewise, the DES Hubble diagram strongly constrains distance-law shape, but the released vector leaves the static law of Eq. (13) nearly degenerate with flat Λ CDM.

The two facts together change the status of the static-universe question. The alternative is no longer the classical proposition “photons lose energy but nothing else changes.” It is a constrained two-channel model: the forward channel must be achromatic, direction-preserving, and cycle-conserving; the non-forward channel must produce the observed dimming without spoiling sharp images and must leave a specific chromatic timing residue.

Wien displacement is useful in this framing because it ties the frequency change to an independently observable spectral-temperature shift. A redshifted thermal sequence is observed at lower effective temperature, so fixed observer bands sweep through different rest-frame parts of the supernova spectral-energy distribution. That effect is expected in both expanding and frequency-shifting descriptions because it is a statement about the measured spectrum. The discriminant is not the existence of the Wien shift; it is whether the complete static transfer operator reproduces the joint spectral, temporal, and luminosity data with the predicted residual chromatic structure.

The present work therefore does not claim that DES has proved a static universe. It establishes a narrower result with a clear experimental consequence: the two canonical supernova signatures often treated as independent proofs of expansion—Hubble-diagram curvature and $(1+z)$ temporal broadening—are not logically unique to metric expansion once a coherent frequency-shifting propagation law is admitted. The released DES data currently leave the distance-law comparison nearly degenerate; the next step is to test the unique chromatic and pipeline-level predictions of the static law.

10 Conclusion

A static-geometry, two-component photon-propagation model can be formulated so that the same coherent frequency shift responsible for redshift also satisfies the DES $b \simeq 1$ temporal-broadening constraint, provided phase cycles are conserved. The energy loss and arrival-rate dilution together generate the effective $(1+z)$ luminosity factor, while a separate $E^{1/2}$ non-forward channel supplies a saturating dimming term and predicts a weak chromatic rise/decay asymmetry.

On the released DES-SN5YR Hubble diagram this one-parameter static law is nearly degenerate with one-parameter flat Λ CDM, $\Delta\chi^2 = +1.1$. The total BBC processing layer changes that comparison by $-11.7 \chi^2$ units relative to a reconstructed pre-BBC diagnostic vector. Existing DES reference-cosmology tests substantially constrain ordinary FLRW conditioning at the amplitude of the present comparison, but they do not replace a matched simulation with the non-FLRW propagation law itself.

The thesis is therefore experimentally specific. The static model survives the released supernova distance and bulk-width tests; it now owes the community two measurements: the predicted observer-frame chromatic timing asymmetry and a matched bias-correction/forward-model rerun. Either can falsify the model. If both succeed, the supernova evidence usually attributed uniquely to accelerating expansion would have a quantitatively viable static propagation alternative.

Data and code availability

The DES-SN5YR light curves, simulations, distance products, and covariance matrices are public. The companion model-discrimination reproduction package is maintained in the companion GitHub

repository. A frozen archive is planned under the reserved DOI 10.5281/zenodo.22022089. The broader theory and cosmology calculations are maintained in the QFD source repository.

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